

In-band Noise Mitigation of Broadband Chirped Signals Based on Joint Time-Frequency Photonic Processing

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Abstract We demonstrate a real-time microwave photonic approach for in-band noise mitigation using time–frequency domain processing. In the experiment, chirped signals with up to 23-GHz bandwidth are recovered under strong optical noise, achieving up to ~9 dB SNR improvements.

Introduction

Broadband waveform processing at microwave and millimeter-wave frequencies is central to modern optical communication, radar, and sensing systems. In particular, chirped signals are widely employed due to their high-resolution ranging capability and efficient spectral utilization [1], [2]. However, their wide spectral occupancy also makes them highly susceptible to noise, which can significantly degrade waveform fidelity and reduce detection accuracy, resolution and overall system performance [3], [4], [5]. This issue becomes especially critical in practical scenarios involving long-distance propagation or weak signal returns, where the useful signal approaches or even falls below the noise floor. A key challenge is the suppression of in-band noise. As this overlap spectrally with the inherently broadband signal and hence cannot be removed using conventional static filtering techniques, in-band noise requires more sophisticated processing strategies to avoid losing the information carried by the target signal. Specifically, the continuously varying instantaneous frequency of a chirped signal implies a strong coupling between its temporal and spectral features, making static filtering approaches ineffective.

Denosing methods based on digital signal processing offers high programmability and fine spectral control [6], but they remain limited for real-time processing of signals with multi-GHz instantaneous bandwidths. Microwave photonics provides a compelling alternative with its inherent advantages in terms of bandwidth and flexibility. Nevertheless, existing photonic filtering techniques are often insufficient for in-band noise suppression along complex, high-speed signals [7], [8], [9]. Coherent spectral manipulation have shown promise, but it is typically constrained to (sub-)nanosecond waveforms [10]. Temporal cloaking-based noise filtering can selectively suppress undesired temporal regions, yet has so far been demonstrated mainly for relatively narrowband (MHz bandwidth) signals [11].

Recently, we have introduced a time-varying

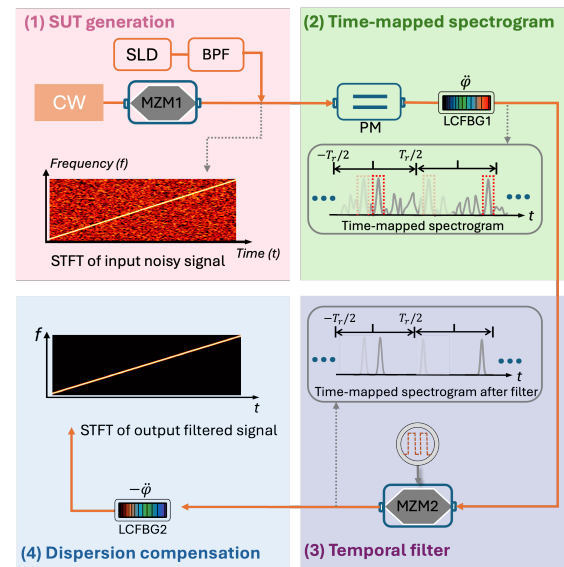


Fig. 1: Schematic of the proposed time-frequency denoising system, CW: continuous-wave light; SLD: super luminescent diode; BPF: band-pass filter; MZM: Mach-Zehnder modulator; PM: phase modulator; LCFBG: linearly chirped fiber Bragg grating; STFT: short-time Fourier transform.

microwave photonic filter scheme based on joint time–frequency (TF) domain manipulation [12], enabling real-time processing of broadband signals with nanosecond-level reconfigurability and MHz-scale frequency resolution. By mapping the short-time Fourier transform (STFT) of the incoming signal into the temporal domain through cascaded phase-modulation and group-velocity dispersion, spectral information is converted into a time-resolved form that can be selectively processed via programmable temporal gating. In this work, we demonstrate the use of this TF-domain processing method for mitigating in-band noise in highly non-stationary signals, such as chirped waveforms. We experimentally validate the approach on broadband microwave chirped signals under different noise conditions. We consider the case of optical noise as this is particularly problematic when the signal is upconverted into the optical domain for efficient distribution or processing, such as in microwave-photonic radar and radio-over-fiber systems [5], [13]. We first

demonstrate the recovery of a 20-GHz-bandwidth linearly chirped signal, achieving an ~ 9 dB signal-to-noise ratio (SNR) improvement in the presence of substantial optical noise. Furthermore, we show the capability of the method to recover a more sophisticated waveform, namely, a 20-GHz double-chirped signal, that is deeply buried in strong optical noise of comparable bandwidth, with a demonstrated SNR enhancement of up to ~ 8.6 dB.

Operation Principle

The proposed TF denoising scheme is illustrated in Fig. 1 and it consists of three main operations: time-frequency mapping; customized temporal modulation for user-defined time-frequency filtering, and processed signal reconstruction. In the experiment, the signal under test (SUT), comprising the target chirped signal, is modulated onto a continuous-wave (CW) optical carrier (1550 nm) using a 40-GHz Mach–Zehnder modulator (MZM). An electronic arbitrary waveform generator (AWG) is used to implement the input microwave signal, the temporal phase and filtering patterns. Note that the modulation produces a double-sideband spectrum; for clarity, only the positive sideband is highlighted in all shown plots. In the experiment, the optical noise is generated from a super luminescent diode (SLD) and filtered by a bandpass filter (BPF) with a bandwidth of 2 nm. The SNR is calculated in the TF domain using the STFT. For each time slice, the frequency bins corresponding to the chirp traces are identified, and their union defines the signal region in the TF plane. The signal power is obtained by summing the spectrogram power within these regions over all time indices. The remaining frequency bins are treated as noise, and the noise power is computed by summing the spectrogram power over these bins across all time indices.

The noisy optical signal is then processed by a time-mapped spectrogram (TM-SP) operation based on the Talbot array illuminator spectrogram design [12]. This process combines a periodic multi-level temporal phase modulation with quadratic spectral phase dispersion, implemented using an electro-optic phase modulator (PM) and a linearly chirped fiber Bragg grating (LCFBG), respectively. Specifically, the TAI phase modulation consists of a periodic pattern and each period is divided into q discrete steps, each with a length of t_s , according to the following expression: $\varphi_n = -\pi n^2 \frac{q-1}{q}$, for $n = 1, 2, \dots, q$ [12]. This pattern is then repeated with a period $T_r = qt_s$. The quadratic spectral phase induced by the LCFBG satisfies $\ddot{\phi} = \frac{qt_s}{2\pi}$ (where $\ddot{\phi}$ is the slope of group delay as a function of the radial frequency variable). Following propagation through this system, the spectral components of

the input signal are mapped along the time domain according to a frequency-to-time mapping relationship, $\Delta\omega_t \rightarrow \Delta t/\ddot{\phi}$, where $\Delta\omega_t$ and Δt are defined relative to the centre of each analysis window, producing a sequence of time-resolved spectral replicas, each extending over each analysis window T_r . This enables direct manipulation of the SUT frequency components in the time domain. A programmable gating pattern is then applied using a second MZM. By aligning the temporal gates with the time–frequency trace of the chirped signal, the instantaneous frequency components of the desired signal can be selectively preserved while suppressing noise. Finally, the filtered signal is reconstructed using a second dispersive element with opposite dispersion, followed by photodetection. The results are recorded with a 28-GHz real-time oscilloscope.

Experimental results

We first experimentally demonstrate isolating a chirped signal from relatively strong optical noise. A broadband linear chirped (0.5 GHz to 20 GHz) spanning over a 135 ns duration is combined with the optical noise, resulting in an input SNR of about -7 dB. The signal is then severely degraded, with the chirped structure nearly indistinguishable in both its temporal and spectral representations, as shown by the numerical STFT in Fig. 2 (a) and the zoom-in plot.

To achieve the target selective filtering, the signal is then transformed into a TM-SP, with parameters $t_s \sim 10.86$ ps ($= 1/92$ GHz), $T_r \sim 0.76$ ns and dispersion $\ddot{\phi} \sim 2,600$ ps², corresponding to a

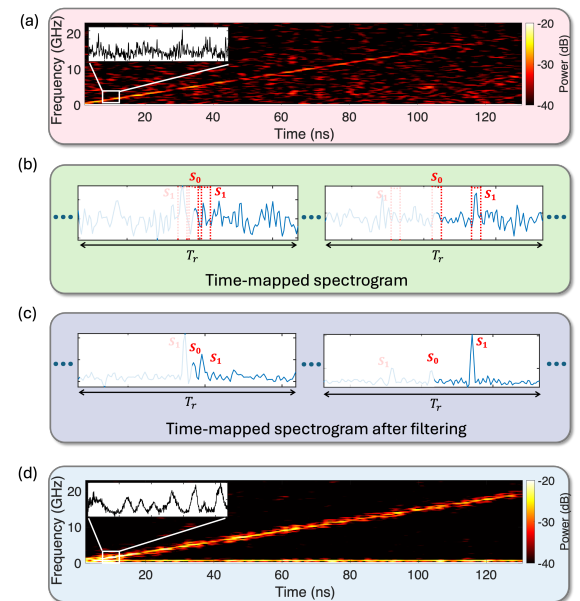


Fig. 2: Experimental results of optical noise suppression. (a) Numerical STFT of the input noisy signal. (b) Zoomed-in TM-SP of the measured output, along with the temporal filtering pattern (dashed red). (c) Zoomed-in TM-SP after filtering. (d) STFT of the recovered microwave signal after denoising.

46 GHz full (double side-band) analysis bandwidth and ~ 1.3 GHz frequency resolution. Figure 2(b) presents two representative zoomed-in regions of the measured TM-SP waveform, each covering a different analysis period T_r . For clarity of visualization, only the positive-frequency components are shown in the figure. The plots clearly show that broadband noise is distributed across the entire temporal window, whereas the chirped signal is mapped into localized pulses (S_1) with an evolving location across successive time frames.

Since the noise and signal components exhibit similar amplitudes in the TM-SP, the filtering mask needs to be designed based on the known chirp profile, which is typically available in radar or other sensing systems. A thresholding device could be alternatively used without prior knowledge of the waveform though this would be limited to operation on moderate noise conditions where the signal spectral peaks remain above the noise background. In the present experiment, the filtering temporal mask consists of a series of rectangular gating windows, each with a duration of t_s , represented as red dotted patterns. The gates are aligned with the mapped signal pulses, denoted as S_1 , within each analysis window to isolate the desired frequency content. Note that to recover the filtered signal in the subsequent step, here the DC component (S_0) also needs to be maintained. The effectiveness of the filtering process is shown in Fig. 2(c), where the TM-SP after temporal gating is presented for the same regions as in Fig. 2(b). A clear strong mitigation of the noise components is observed, while the signal features become more prominent and well-defined. Finally, the numerically calculated STFT of the recovered filtered waveform after the final dispersion compensation and photodetection stages, shown in Fig. 2(d), demonstrates that the chirped structure is successfully restored. The output SNR is estimated to be ~ 2 dB, corresponding to an SNR improvement of ~ 9 dB.

We further evaluate the system for denoising a more sophisticated waveform. For this purpose, the system is reconfigured with $t_s \sim 21.74$ ps and a period length of $T_r \approx 1.5$ ns, using the same dispersion setup. The SUT is a double-chirped signal spanning DC - 20 GHz over 135 ns. This is again corrupted by strong broadband optical noise such that the double-chirped signal is completely buried in noise, making it impossible to identify the underlying signal structure. Fig. 3(a) shows the numerical STFT of the detected noisy input signal (SNR ~ -7.2 dB). Fig. 3(b) shows two zoomed-in regions of the measured TM-SP of the SUT, each spanning a duration of T_r . The spectrogram clearly indicates the presence of strong noise, making the chirped signal components

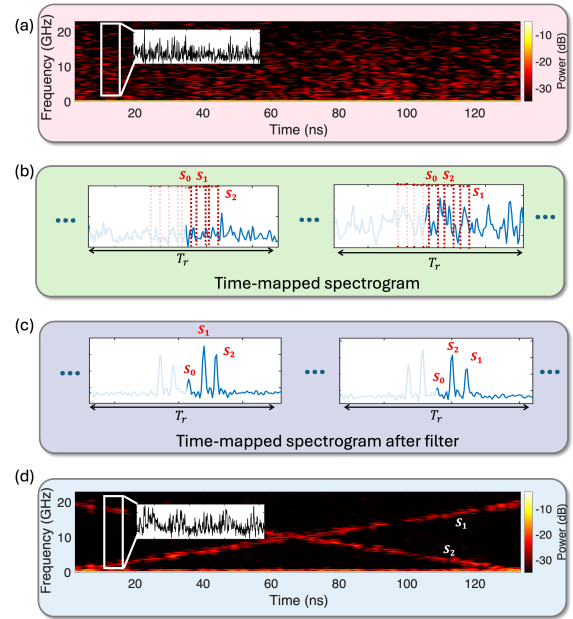


Fig. 3: Experimental results of strong noise suppression. (a) Numerical STFT of the input noisy signal. (b) Zoomed-in TM-SP of the measured output, along with the temporal filtering pattern (dashed). (c) Zoomed-in TM-SP after filtering. (d) STFT of the recovered microwave signal after denoising.

difficult to distinguish. A temporal filtering mask is then applied to select the instantaneous frequency components of the two chirps, labelled S_1 and S_2 , as well as the DC part (S_0), as illustrated by the red dashed curves, each gating the desired pulse with a duration equal to t_s . The TM-SP after filtering is shown in Fig. 3(c), using the same zoomed-in regions as in Fig. 3(b). Finally, the filtered signal is reconstructed after the second dispersive stage. The corresponding numerical STFT in Fig. 3(d) confirms that the noise has been strongly mitigated while preserving the double-chirped signal, with an output SNR of ~ 1.4 dB, corresponding to an improvement of ~ 8.6 dB.

Conclusion

In summary, we have demonstrated a real-time microwave photonic approach for mitigating in-band noise in broadband chirped signals based on joint TF manipulation. Significant SNR improvements have been achieved. While the reported results focus on the mitigation of optical noise, the same framework is directly applicable to microwave-domain noise suppression, as confirmed by recent experimental results.

Overall, the proposed framework provides a promising pathway toward robust signal recovery in noisy challenging environments, and it should be particularly interesting for denoising highly non-stationary broadband signals from the microwave to the optical domains, such as those found in remote sensing (radar and lidar) and advanced communication (e.g., frequency hopping) systems.

Acknowledgements

We would like to thank the support from NSERC (Canada) and FRQNT (Québec).

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